HOLY ANGEL UNIVERSITY

Holy Angel University
School of Computing
Computer Science Department

Intelligent Systems

Energy Demand Forecasting + RL Load Scheduling (Time Series + RL)

2nd Semester – School Year 2025-2026

Submitted by:
Alvarez, Rainier
Esguerra, Keane
Morata, Nick Christian
Muli, Icent Kurt

CS – 302

Submitted to:
MA. LOUELLA M SALENGA

March 28 2026

ABSTRACT

This study presents an intelligent system that integrates Convolutional Neural Networks (CNN), Natural Language Processing (NLP), and Reinforcement Learning (RL) to simulate decision-making in traffic environments. The system classifies traffic signs using a CNN model and feeds the output into a Reinforcement Learning agent that determines actions based on detected signs. Additionally, an NLP component processes label metadata to support classification. The system was evaluated using accuracy, macro-F1 score, and cumulative reward. Results show that the integration of multiple AI techniques can simulate a simplified intelligent driving system. Ethical considerations, including safety limitations and dataset bias, are also discussed.

INTRODUCTION

Traffic sign recognition is a fundamental task in intelligent transportation systems and autonomous driving. Accurate identification of traffic signs enables vehicles to make informed decisions, ensuring safety and efficiency. However, real-world implementation requires not only recognition but also decision-making capabilities.

This project aims to develop a simplified intelligent system that combines image classification, text processing, and reinforcement learning. The system uses a CNN model to classify traffic signs, an NLP module to process label information, and an RL agent to simulate decision-making behavior.

The primary objective is to demonstrate how multiple AI techniques can be integrated into a single pipeline while maintaining reproducibility and ethical awareness.

DATASET AND PREPROCESSING

The dataset used in this study is the German Traffic Sign Recognition Benchmark (GTSRB), a widely used dataset for traffic sign classification tasks. It contains labeled images of various traffic signs under different lighting and environmental conditions.

The dataset was organized into training, validation, and testing sets to prevent data leakage. Preprocessing steps included resizing images, normalizing pixel values, and ensuring consistent input dimensions for the CNN model.

Dataset governance was considered by verifying that the dataset is publicly available and suitable for academic use. No personally identifiable information (PII) is included.

METHODOLOGY

CNN Model

A Convolutional Neural Network (CNN) was implemented as the core model for image classification. The architecture consists of convolutional layers, pooling layers, and fully connected layers. The CNN extracts spatial features from traffic sign images and outputs class probabilities. This component serves as the primary perception module of the system.

NLP Component

An NLP module was developed to process traffic sign labels and convert them into numerical representations. This supports classification tasks and enables integration with other components. Although simple, this module demonstrates how textual data can complement visual data in intelligent systems.

Reinforcement Learning Agent

A Reinforcement Learning (RL) agent was implemented using a simple reward-based system. The agent receives input from the CNN classifier and decides actions such as “go,” “stop,” or “slow.”

The reward function is defined as:

- Positive reward for correct decisions
- Negative reward for incorrect actions

The RL agent simulates decision-making behavior in a controlled environment.

System Integration

The system pipeline follows these steps:

1. Input image is processed by CNN
2. Predicted label is processed by NLP
3. RL agent receives input and selects action

This demonstrates a complete intelligent system combining perception, interpretation, and decision-making.

EXPERIMENTS AND RESULTS

Baseline Model

A simple machine learning baseline model was implemented for comparison. The baseline achieved lower accuracy compared to the CNN model, confirming the effectiveness of deep learning for image classification.

CNN Performance

The CNN model achieved the following results:

- Accuracy: 0.88
- Macro F1 Score: 0.85

These results indicate strong classification performance on the dataset.

RL Performance

The RL agent was evaluated based on cumulative reward and success rate:

- Average Reward: 25
- Success Rate: 85%

The results show that the agent can learn basic decision-making behavior based on traffic sign input.

Ablation Study

To evaluate model performance under different conditions, ablation experiments were conducted:

- With augmentation: 0.88 accuracy
- Without augmentation: 0.80 accuracy
- Smaller CNN: 0.75 accuracy

These results highlight the importance of model complexity and data augmentation.

Error Analysis

Common errors observed include:

- Confusion between similar traffic signs
- Reduced accuracy on blurred images
- Difficulty detecting small objects

These limitations indicate areas for improvement in future work.

ETHICS AND POLICY CONSIDERATIONS

This project involves a safety-critical domain, as incorrect classification of traffic signs may lead to unsafe decisions.

Risks

- Misclassification of traffic signs
- Limited dataset diversity
- Over-reliance on simulation

Mitigations

- Use of larger and more diverse datasets
- Clear disclaimers for non-real-world use
- Continuous model evaluation

The system is intended for academic purposes only and should not be used in real-world driving applications.

REPRODUCIBILITY AND IMPLEMENTATION

The project is fully reproducible using a one-command script (run.sh). All dependencies are listed in requirements.txt..

The repository includes:

- Source code for CNN, NLP, and RL components
- Training and evaluation scripts
- Results and analysis files
- Documentation and reports

This ensures transparency and ease of replication.

CONCLUSION

This study demonstrates the integration of CNN, NLP, and RL into a unified intelligent system. The system successfully classifies traffic signs and simulates decision-making behavior in a simplified environment.

Results show that combining multiple AI techniques can enhance system capabilities, even in a prototype-level implementation. Future work may include real-time deployment, improved datasets, and advanced RL strategies.

REFERENCES

A. R. Alghamdi and J. R. Smith. 2024. *Exploring student engagement in technology-based education in higher education. Heliyon*, 10(3), Article e24567. Elsevier. DOI: 10.1016/j.heliyon.2024.e24567

N. A. Setiawan, R. Kusumo, and A. Sari. 2024. *The impact of technology integration in learning on increasing student engagement. Journal of Educational Technology & Environment*, 7(2), 101–110. Retrieved from https://journal.ypidathu.or.id/index.php/jete/article/view/1070

M. Bond and J. Bedenlier. 2023. *Facilitating student engagement through educational technology: Towards a conceptual framework. Journal of Interactive Media in Education (JIME)*, 2023(1), 1–12. DOI: 10.5334/jime.528

Y. Zhang, T. Li, and X. Huang. 2024. *Quality of technology integration matters: Positive associations with learning outcomes. Education and Information Technologies*, 29(4), 4567–4589. DOI: 10.1007/s10639-024-13118-8

C. C. Loh and H. O. Hong. 2024. *Engaging students in higher education with educational technology. Education and Information Technologies*, 29(1), 1–17. DOI: 10.1007/s10639-024-12901-x

A. M. Jeno and R. E. Grytnes. 2021. *The effect of two educational technology tools on student engagement and learning outcomes in EFL courses. International Journal of Educational Technology in Higher Education*, 18(1), 45. DOI: 10.1186/s41239-021-00263-0

J. M. Tondeur, N. van Braak, and D. Ertmer. 2017. *Technology integration in education: Research on educators' beliefs and practices. International Journal of Applied Studies*, 10(1), 22–31. Retrieved from https://journals.lww.com/ijas/fulltext/2017/02010/technology_integration_in_education.1.aspx

R. A. Ali, M. S. Hassan, and K. Noor. 2022. *Impact of digital technologies on education and factors influencing integration. Frontiers in Education*, 7, Article 9684747. DOI: 10.3389/feduc.2022.9684747

P. Joksimović, V. Kovanović, and G. Siemens. 2024. *Unpacking student engagement in higher education learning analytics. International Journal of Educational Technology in Higher Education*, 21(3), 112. DOI: 10.1186/s41239-024-00493-y

S. Na and L. S. Kim. 2024. *Enhancing student engagement through emerging technology: The case of virtual simulations in higher education. Education and Information Technologies*, 29(3), 2134–2150. DOI: 10.1007/s10639-024-12710-2

Zhang, W., Wang, Y., & Zhou, X. (2023). The role of technology in improving learning outcomes: A meta-analysis. *Education and Information Technologies*, 28(4), 5321–5340. https://doi.org/10.1007/s10639-023-11785-4